

EDUCATION

- **University of Colorado Boulder** Boulder, Colorado
Bachelor of Science Aerospace Engineering and Sciences, Minor: Computer Science *Expected 2027*
Relevant courses: Computer Systems, Discrete Mathematics, Experimental and Computational Methods in MATLAB, Data Structures, Algorithms, Orbital Dynamics

WORK

- **Small Satellite Deployable Booms, Undergrad Capstone Project** Boulder, CO
Command and Data Handling Engineer *Aug 2026 - Present*
 - Currently developing a boom structure and payload to enable electronic field measurements on a small satellite in LEO below a noise threshold from spacecraft generated noise.
 - Produced the specific functional system requirements for computer communication, boom deployment and operation, data processing, and packetization for transmission.
- **Office of Information Technology, University of Colorado Boulder** Boulder, CO
Lead IT Analyst *Oct 2022 - Present*
 - Gained expertise in business technology and operations, operating systems, and front-end web development in HTML and CSS.
 - Maintained and updated knowledge base documentation, translating complex technical data into user-friendly instructions, reducing reliance on IT assistance.
 - Collaborated with multiple IT teams to resolve 1000+ software, database, and network issues over a 2-week period.
- **NASA Lunabotics Competition, Oakton Community College** Skokie, IL
Project Manager *Aug 2021 - Aug 2022*
 - Designed, tested, and fabricated an excavation robot to compete against 49 universities in a simulated lunar environment.
 - Designed a new regolith dumping mechanism using one less motor and a physical brake, requiring less wiring and programming.

PROJECTS & EXPERIENCE

- **Electronics and Communications Lab Signals and Communication Processing in MATLAB:** Developed a signal-analysis script to characterize amplitude-modulated waveforms using oscilloscope measurements and MATLAB. Gathered experimental data, implemented FFT-based frequency-domain analysis in MATLAB, and identified dominant carrier and modulation frequency components from measured spectra.
- **Electronics and Communications Lab Frequency Domain and Spectrum Analyzer in MATLAB:** Developed a signal-processing workflow to characterize the response of a sine wave passing through an RC Low-Pass filter. Collected oscilloscope measurements, processed sampled waveform data in MATLAB using FFT analysis, to plot and estimate signal frequency and amplitude characteristics for comparison between time and frequency-domain behavior.
- **MATLAB .png Image Editor:** Developed an image-processing program in MATLAB that decomposed imported images into RGB pixel matrices and applied custom transformation functions for color manipulation and image resizing.
- **C++ Social Media Database Restructuring:** Implemented a C++ data management system using a chained hash table and binary search trees, dynamically organizing user profiles and time-indexed posts while handling hash collisions, pointer-based structures, and dynamic memory allocation.
- **C++ Data Structures and Algorithms:** Implemented a C++ graph-processing program that converted 2D map data into an adjacency-list representation and applied Depth-First Search to identify connected nodes, incorporating file parsing, pointer-based structures, and undirected graph traversal.

TECHNICAL SKILLS

- **Programming Languages:** MATLAB, Python, C++, C, R, HTML, CSS, Node.JS
- **Business & Collaboration Tools:** Microsoft Office Suite (Excel, Word, PowerPoint), GitHub, JIRA, Cisco Platforms, Salesforce, Confluence, Zoom